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Branding is
what people
say about
you when
you are not
in the room.



-Jeff Bezos

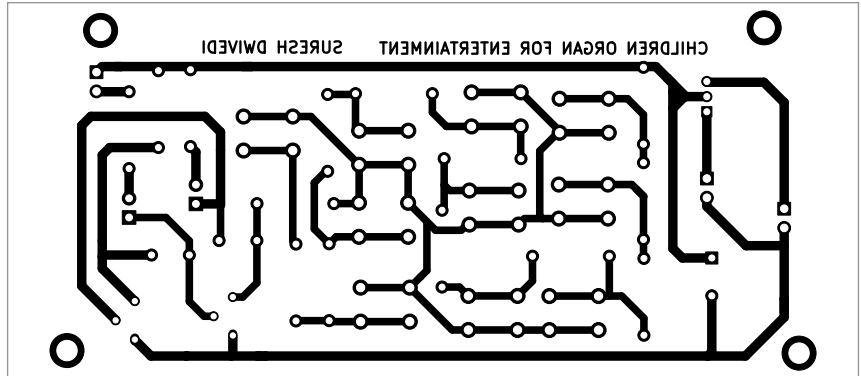


Fig. 3: Actual-size PCB

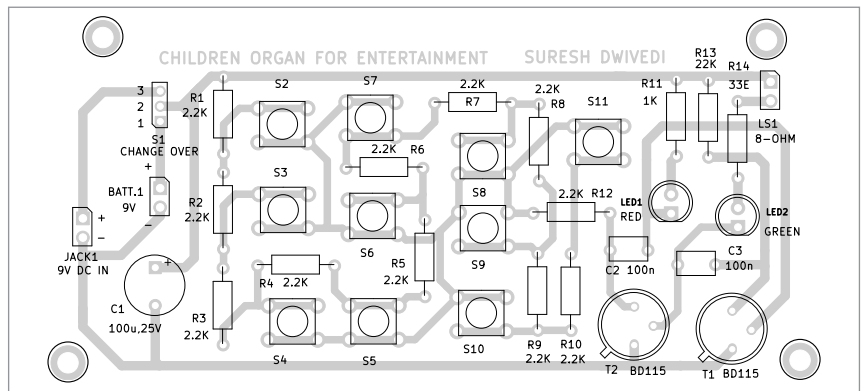


Fig. 4: Components side

a distinct sound, mimicking a simple keyboard or organ. This device offers an effective method for developing an early interest in electronics and music among children.

Construction and testing

An actual-size, single-sided PCB layout for the organ is shown in Fig. 3, with its component layout in Fig. 4. After assembling the circuit on the PCB, the device should be enclosed in a suitable box, positioning all the switches, including the LEDs, on the top of the box to ensure easy access for children. JACK1 and the changeover switch S1 should be mounted on the front side of the box. The speaker should be placed at the rear side of the box, ensuring that five to ten holes are provided for sound output. **EFY**

PARTS LIST

Semiconductors:

- T1, T2 - BD115 NPN transistor
- LED1, LED2 - 5mm LED (red and green)

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R10, R12 - 2.2k Ω
- R11 - 1k Ω
- R13 - 22k Ω
- R14 - 33 Ω

Capacitors:

- C1 - 100 μ F, 25V electrolytic
- C2, C3 - 100nF ceramic disc

Miscellaneous:

- JACK1 - DC socket
- LS1 - 8 Ω , 0.5-watt speaker
- BATT.1 - 9V battery
- S1 - SPST switch
- S2-S11 - Push-to-on switch

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-make-childrens-musical-organ>

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